



BANNARI AMMAN INSTITUTE OF TECHNOLOGY

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22MA201 –ENGINEERING MATHEMATICS II

UNIT 1 – LINEAR FIRST ORDER DIFFERENTIAL EQUATIONS

UNIT 1 & LM 3 – LEIBNITZ’S EQUATION

1 INTRODUCTION

1.1 Leibnitz’s Linear differential equation:

A standard differential equation, introduced by a German mathematician Gottfried Wilhelm Leibniz (or Leibnitz) is called the Leibnitz’s linear differential equation.

Leibnitz introduced a standard form linear differential equation of the first order and first degree.

$$\frac{dy}{dx} + Py = Q$$

It is defined in terms of two variables x and y. In this equation, P and Q are the functions in terms of a variable x.

Gottfried Wilhelm Leibnitz derived a solution for this first order linear differential equation. Hence, this differential equation is called the Leibnitz’s linear differential equation or Leibniz’s linear differential equation. The following equation is the solution for the Leibnitz’s linear differential equation.

$$ye^{\int Pdx} = \int Qe^{\int Pdx} dx + C$$

On both sides of the solution, there is a factor in integration form and it is $e^{\int Pdx}$. It is called the integrating factor, simply written as I.F.

Hence, the solution of the Leibniz’s linear differential equation is also written as follows.

$$y(I.F) = \int Q(I.F)dx + C$$

1.2 Leibnitz’s equation Problems:

1. Solve $y' + 3y = 2x - 1$

Solution:

The standard form of Leibnitz equation is

$$\frac{dy}{dx} + Py = Q$$

Therefore,

$$P(x) = 3, Q(x) = 2x - 1$$

Since the Integrating factor is $I.F = e^{\int P dx}$

$$I.F = e^{\int 3 dx} = e^{3x}$$

To find $y(x)$: $y(I.F) = \int Q(I.F)dx + C$

$$ye^{3x} = \int (2x - 1)e^{3x} dx + C$$

$$\Rightarrow ye^{3x} = \int (2x - 1)e^{3x} dx + C$$

$$\Rightarrow ye^{3x} = (2x - 1)\frac{e^{3x}}{3} - \int 2 \cdot e^{3x} dx + C$$

$$\Rightarrow ye^{3x} = (2x - 1)\frac{e^{3x}}{3} - \frac{2}{9} e^{3x} + C$$

$$\Rightarrow y = (2x - 1)\frac{1}{3} - \frac{2}{9} + \frac{C}{e^{3x}}$$

$$\Rightarrow y = \frac{2x}{3} - \frac{5}{9} + Ce^{-3x}$$

2. Solve $x\left(\frac{dy}{dx} + y\right) = 1 - y$

Solution:

The standard form of Leibnitz equation is

$$\frac{dy}{dx} + Py = Q$$

Since,

$$x\left(\frac{dy}{dx} + y\right) + y = 1$$

$$\Rightarrow x \frac{dy}{dx} + xy + y = 1$$

$$\Rightarrow x \frac{dy}{dx} + y(x + 1) = 1$$

$$\Rightarrow x \frac{dy}{dx} + y(x+1) = 1$$

$$\Rightarrow \frac{dy}{dx} + \left(\frac{x+1}{x}\right)y = \frac{1}{x}$$

$$\text{Therefore } P(x) = \frac{x+1}{x}, Q(x) = \frac{1}{x}$$

Since the Integrating factor is $I.F = e^{\int P dx}$

$$\int P dx = \int \frac{x+1}{x} dx = \int \left(1 + \frac{1}{x}\right) dx = x + \log \log x = \log \log e^x + \log \log x = \log \log (xe^x)$$

$$\text{Therefore, } I.F = e^{\int P dx} = e^{\log(xe^x)} = xe^x$$

To find $y(x)$ using the equation $y(I.F) = \int Q(I.F) dx + C$.

$$y \cdot xe^x = \int xe^x \cdot \frac{1}{x} dx + C$$

$$\Rightarrow y \cdot xe^x = \int e^x dx + C = e^x + C$$

$$\Rightarrow yxe^x = e^x + C$$

$$\Rightarrow xy = 1 + Ce^{-x}$$

$$3. \text{ Solve } (x+1) \frac{dy}{dx} - y = e^{3x} (x+1)^2$$

Solution:

Dividing throughout the equation by $(x+1)$, we get

$$\frac{dy}{dx} - \frac{y}{(x+1)} = e^{3x} (x+1), \text{ which is in the form of Leibnitz's equation, } \frac{dy}{dx} + Py = Q$$

where

$$P = \frac{-1}{(x+1)}, Q = e^{3x} (x+1)$$

$$\text{The integrating factor is } I.F = e^{\int P dx} = e^{\int \frac{-1}{(x+1)} dx} = e^{-\log(x+1)} = e^{\log \frac{1}{(x+1)}} = \frac{1}{(x+1)}$$

∴ The solution is $y(I.F) = \int Q(I.F)dx + C \Rightarrow$

$$y\left(\frac{1}{(x+1)}\right) = \int e^{3x}(x+1)\left(\frac{1}{(x+1)}\right)dx + C$$

$$\Rightarrow y\left(\frac{1}{(x+1)}\right) = \int e^{3x}dx + C \Rightarrow y\left(\frac{1}{(x+1)}\right) = \frac{e^{3x}}{3} + C \Rightarrow y = \frac{e^{3x}(x+1)}{3} + C(x+1)$$

4. Solve $\left(\frac{e^{-2\sqrt{x}}}{\sqrt{x}} - \frac{y}{\sqrt{x}}\right)\frac{dy}{dx} = 1$

Solution:

$\frac{dy}{dx} = \frac{e^{-2\sqrt{x}}}{\sqrt{x}} - \frac{y}{\sqrt{x}} \Rightarrow \frac{dy}{dx} + \frac{1}{\sqrt{x}}y = \frac{e^{-2\sqrt{x}}}{\sqrt{x}}$, which is in the form of Leibnitz's equation, where

$$P = \frac{1}{\sqrt{x}}, Q = \frac{e^{-2\sqrt{x}}}{\sqrt{x}}$$

The integrating factor is $I.F = e^{\int Pdx} = e^{\int \frac{1}{\sqrt{x}}dx} = e^{2\sqrt{x}}$

∴ The solution is $y(I.F) = \int Q(I.F)dx + C \Rightarrow y(e^{2\sqrt{x}}) = \int \frac{e^{-2\sqrt{x}}}{\sqrt{x}}(e^{2\sqrt{x}})dx + C$

$$ye^{2\sqrt{x}} = 2\sqrt{x} + C \Rightarrow y = (2\sqrt{x} + C)e^{-2\sqrt{x}}$$

$$\therefore \text{Ans } y = (2\sqrt{x} + C)e^{-2\sqrt{x}}$$

5. Solve $x \log \log x \frac{dy}{dx} + y = \log \log x^2$

Solution:

$\frac{dy}{dx} + \frac{y}{x \log x} = \frac{2 \log x}{x \log x}$, which is in the form of Leibnitz's equation, where

$$P = \frac{1}{x \log x}, Q = \frac{2 \log x}{x \log x}$$

The integrating factor is $I.F = e^{\int Pdx} = e^{\int \frac{1}{x \log x}dx} = e^{\int \frac{1}{u}du} = e^{\log u} = u = \log x$ Put

$$\log x = u \Rightarrow \frac{1}{x}dx = du$$

∴ The solution is $y(I.F) = \int Q(I.F)dx + C \Rightarrow$

$$y(\log x) = \int \frac{2}{x} (\log x) dx + C = 2 \int u du + C = 2 \left(\frac{u^2}{2} \right) + C = (\log x)^2 + C$$

$$\therefore \text{Ans } y = \log x + \frac{C}{\log x}$$

6. Solve $y \log \log y dx + (x - \log \log y) dy = 0$

Solution:

$$\frac{dx}{dy} + \frac{x}{y \log y} = \frac{\log y}{y \log y} = \frac{1}{y}, \text{ which is in the form of Leibnitz's equation, where}$$

$$P = \frac{1}{y \log y}, Q = \frac{1}{y}$$

$$\text{The integrating factor is } I.F = e^{\int P dy} = e^{\int \frac{1}{y \log y} dy} = e^{\log(\log y)} = \log y$$

$$\therefore \text{ The solution is } x(I.F) = \int Q(I.F) dy + C \Rightarrow x(\log y) = \int \frac{1}{y} (\log y) dy + C$$

$$\Rightarrow x(\log y) = \frac{(\log y)^2}{2} + C$$

$$\therefore \text{Ans } x = \frac{\log y}{2} + \frac{C}{\log y}$$

7. Solve $\frac{dy}{dx} - \frac{2y}{x+1} = (x+1)^3$

Solution:

$$\frac{dy}{dx} - \frac{2y}{(x+1)} = (x+1)^3, \text{ which is in the form of Leibnitz's equation, where}$$

$$P = -\frac{2}{(x+1)}, Q = (x+1)^3$$

$$\text{The integrating factor is } I.F = e^{\int P dx} = e^{\int \frac{-2}{(x+1)} dx} = e^{-\log(x+1)^2} = e^{\log \frac{1}{(x+1)^2}} = \frac{1}{(x+1)^2}$$

$$\therefore \text{ The solution is } y(I.F) = \int Q(I.F) dx + C \Rightarrow$$

$$y \left(\frac{1}{(x+1)^2} \right) = \int (x+1)^3 \left(\frac{1}{(x+1)^2} \right) dx + C$$

$$\Rightarrow y \left(\frac{1}{(x+1)^2} \right) = \int (x+1) dx + C \Rightarrow y \left(\frac{1}{(x+1)^2} \right) = \frac{(x+1)^2}{2} + C$$

$$\Rightarrow \text{Ans } y = \frac{(x+1)^4}{2} + C(x+1)^2$$

Practise Problems:

1. $\frac{dy}{dx} + \frac{y}{x} = x^2$
2. $x \frac{dy}{dx} + y = x e^x$
3. Solve $y' + 5y = 4x - 1$

Discussion questions:

1. For what purpose Leibnitz equation is used?
2. What are the techniques adopted in solving the given differential equations?
3. Can you able to solve the differential equations with product of two or more functions?